Subject Description Form

Subject Code	COMP 5311
Subject Title	Internet Infrastructure and Protocols
Credit Value	3
Level	5
Pre-requisite/ Exclusion	Nil
Objectives	The overall objective of this course is to build up a solid understanding on the networking technologies underpinning the current Internet infrastructure. This course would serve as an important pre-requisite for other more advanced topics, such as network security, network measurement and diagnosis, wireless and mobile networks, and multimedia networking. The teaching approach will be based on in-depth problem-solving and hands-on class projects. Specifically, 1. understand the TCP/IP technology underpinning Internet; 2. understand the original design philosophy of Internet, and the strength and weaknesses of the then designed Internet in today's computing environment; 3. explore some most up-to-date development in the Internet technology; and 4. acquire knowledge in one specific Internet topic through a group project.
Intended Learning Outcomes	After completing the subject, students should be able to: a) demonstrate critical thinking and in-depth understanding of specialized technical journals and research articles in the professional computer networking domain; b) utilize various specialized network diagnosis tools and apply coherent and advanced body of networking knowledge to conduct in-depth research on network protocols develop creative solutions, and conduct performance evaluations in practical settings; c) critically review and summarize latest advances in specialized networking topics that require foundational in-depth understanding of network protocols such as TCP/IP suite, leveraging emerging technologies for global challenges.
Subject Synopsis/ Indicative Syllabus	• Data-link networks and IP: shared medium and point-to-point networks; the internetworking problem, the hour-glass model, address resolution, IP fragmentation, packet reordering, IP addressing. • IP forwarding: longest prefix match algorithms, routing vs switching, IP address lookup, packet classification, IP tunnelling, ICMP. • End-to-end issues and protocols: end-to-end argument, end-to-end reliability, TCP and UDP, sliding window protocol, acknowledgment strategies.

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Control congestion in Internet: TCP slow-start and congestion avoidance, TCP fast retransmit and recovery, fairness, buffer management, packet scheduling, and queue management.
Applications protocols, e.g., DNS and HTTP, and their interactions with the lower layers.
Internet routing: Internet topology, distance vector, link state, and path vector routing protocols, convergence and routing loops, Routing Information Protocol, Open Shortest Path Protocol, Border Gateway Protocol, Inter-AS relationship.
Design philosophy of IP and TCP, and future challenges.

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<tr><td>Teaching/Learning Methodology</td><td>Class activities including - lecture, tutorial, lab, workshop seminar where applicable</td></tr>
<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>
<table>
<tr><td rowspan="2">Specific Assessment Methods/Tasks</td><td rowspan="2">% weighting</td><td colspan="3">Intended subject learning outcomes to be assessed</td></tr>
<tr><td>a</td><td>b</td><td>c</td></tr>
<tr><td>Assignments, Tests & Projects</td><td>30</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>Final Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>Total</td><td>100</td><td colspan="3"></td></tr>
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<tr><td>Student study effort expected</td><td>
<table>
<tr><td colspan="2">Class Contact:</td></tr>
<tr><td>Class activities (lecture, tutorial, lab)</td><td>39 hours</td></tr>
<tr><td colspan="2">Other student study effort:</td></tr>
<tr><td>Assignments, Quizzes, Projects, Exams</td><td>66 hours</td></tr>
<tr><td>Total student study effort</td><td>105 hours</td></tr>
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<tr><td>Reading list and references</td><td>
(1) J. Kurose and K. Ross, Computer Networking: A Top-Down Approach, 8th Edition, Pearson, 2020.
(2) L. Peterson and B. S. Davie, Computer Networks: A Systems Approach, https://book.systemsapproach.org/, 2019.
(3) Academic Journals and Conference Papers such as publications in IEEE/ACM Transactions on Networking, IEEE Transactions on Mobile Computing, ACM SIGCOMM, ACM MobiCom, and IEEE INFOCOM.
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